# Week 3 BA

## 2024 July07 Quiz-1

**Comprehension for questions 1-3** (source Q17)

"Bro" has always had a passion for data and the science behind it. Hence, to gain a formal education, Bro has explored and identified three different certificate programme provides: "The IITM BSc Program", "IITM NPTEL Analytics Certification" and "A Random Dude Certification". Before, opting for one, Bro has decided to identify the success of the program providers in terms of job opportunities after completion. The success is measured in three levels, "Level-1: Placement with High Package", "Level-2: Placement with Low Package" and "Level-3: No placement". From past data, it is seen that of all students who completed the "The IITM BSc Program" 30% of past students are in Level-1 and 45% are in Level-2. Similarly, of all students who completed the "IITM NPTEL Analytics Certification", 45% are in Level-1 and 30% are in Level-2. Finally, of all students who completed the "A Random Dude Certification", 25% are in the Level-1 and the rest are in Level-3. Based on past admission trends, it is seen that 40% of the student population are in the "The IITM BSc Program", 30% are in the "IITM NPTEL Analytics Certification" and the remaining have opted for "A Random Dude Certification". Assuming the necessary mutual exclusiveness criteria, answer the given sub questions.

**1.** [Source Q18; MCQ]
If Bro wants to be a Level-1 student, then which certification program should he choose?

(a) The IITM BSc Program

(b) IITM NPTEL Analytics Certification

(c) A Random Dude Certification

(d) Cannot answer, as I do not know how to solve the problem

[Source ans: b]

**2.** [Source Q19; SA]
If a total of 1000 students are taken (across all three programmes) then how many will be in "Level-2"? (Note: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")

[Source ans: 770]

**3.** [Source Q20; MCQ]
Bro finds Dr. Milo (a poor student and a very strict BA course instructor) who is not placed, then which program would you suspect Dr.Milo has completed?

(a) The IITM BSc Program

(b) IITM NPTEL Analytics Certification

(c) A Random Dude Certification

(d) Cannot answer, as Dr. Milo is too dumb to even qualify for any of these

[Source ans: c]

**Comprehension for questions 4-7** (source Q21)

Competitive eating a very popular sport in Western Countries. Competitive eating, or speed eating, is a sport in which participants compete against each other to eat large quantities of food, usually in a short time period. There is a general myth that people who have larger waist sizes eat faster and hence have a better chance of winning such competitions. Hence, to check if this is a dependence between waist size and eating speed the data (as specified in Table-3) was collected from interested and willing participants. In this data, the time to complete a A2B Family Dosa (which is equivalent to ~4 Dosas) was considered. The time was grouped into three categories as (i) Fast (less than 2 mins), (ii) Medium (greater than 2 mins but less than 4 minutes) and (iii) Slow (a time greater than 4 minutes). The participants who took part in the study where categorised into five categories based on their waist size as (i) S, (ii) M, (iii) L, (iv) XL, (v) XXL. The entry in a cell indicates the number of participants who completed the dosa in the specified time bucket and have the indicated waist size. Given this information,answer the subquestions.

| *Waist Size* | *Eating Speed* | | |
|---|---|---|---|
| | **Fast** | **Medium** | **Slow** |
| **S** | 20 | 27 | 23 |
| **M** | 13 | 22 | 19 |
| **L** | 13 | 19 | 12 |
| **XL** | 20 | 27 | 22 |
| **XXL** | 18 | 24 | 21 |

Table-3

**4.** [Source Q22; SA]
How many participants with a waist size of "L" are expected to complete the Family Dosa with "Medium" speed? (Note: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")

[Source ans: 17 to 18]

**5.** [Source Q23; SA]
What is the value of the computed test statistic used to perform the desired hypothesis test? (Note: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")

[Source ans: 0.5 to 1.5]

**6.** [Source Q24; SA]
How many degrees of freedom is present for the test statistic, used to perform the desired hypothesis test

[Source ans: 8]

**7.** [Source Q25; MSQ]
If the p-value for the hypothesis test is 0.17, then which of the following statements are TRUE (choose all that is applicable)?

(a) At a 5% level of significance, conclude that there is DEPENDENCE between thewaist size and eating speed

(b) At a 5% level of significance, conclude that there is INDEPENDENCE betweenthe waist size and eating speed

(c) At a 25% level of significance, conclude that there is DEPENDENCE betweenthe waist size and eating speed

(d) At a 25% level of significance, conclude that there is INDEPENDENCE betweenthe waist size and eating speed

(e) None of these are TRUE

[Source ans: b, c]

# 2024 Sep01 ET

**Comprehension for questions 8-10** (source Q11)

The weekly sales in 4 retail stores selling 3 styles of jeans are given in Table-1. The aim is to see if the sales of jeans (jeans styles) is independent across the different stores. Then answer the given subquestions.

| Jeans Style | Number of jeans of a particular style sold in a week at store | | | |
|---|---|---|---|---|
| | 1 | 2 | 3 | 4 |
| A | 72 | 70 | 112 | 43 |
| B | 80 | 76 | 114 | 54 |
| C | 50 | 45 | 89 | 30 |

Table-1

**8.** [Source Q12; SA]
What is the expected number of jeans of Style-B that would be sold in Store-3? (Note: Input your answer rounded to two decimal places. For example, if your answer is "1.235" then enter the answer as "1.24")

[Source ans: 121 to 123]

**9.** [Source Q13; SA]
How many degrees of freedom does are present in the hypothesis test conducted for this problem? (Note: Input your answer rounded to two decimal places. For example, if your answer is "1.235" then enter the answer as "1.24")

[Source ans: 6]

**10.** [Source Q14; MCQ]
If the tabulated value for the test statistic is 5.78 and the computed value of the test statistic is 2.7, then which of the following statements are TRUE

(a) Reject the Null and conclude that the sales of jeans (jeans styles) acrossdifferent stores is independent

(b) Reject the Null and conclude that the sales of jeans (jeans styles) acrossdifferent stores is NOT independent

(c) Do not reject the Null and conclude that the sales of jeans (jeans styles) acrossdifferent stores is NOT independent

(d) Cannot say as the alternative hypothesis is not specified

(e) None of these

[Source ans: e]

**Comprehension for questions 11-13** (source Q15)

DMS lighting produces 3 types of bulbs (Type-A, Type-B and Type-C). The probability that a Type-A bulb will give more than 100 hours (>= 100 hours) of life is 0.7, with the corresponding probabilities of Type-B and Type-C bulbs being 0.4 and 0.3 respectively. In any production, 20% of the bulbs are of Type-A, 30% are of Type-B and 50% are of Type-C. Then answer the given subquestions

**11.** [Source Q16; SA]
What is the probability that a bulb from the production lot will give a life of more than 100 hours? (Note: Input your answer rounded to two decimal places. For example, if your answer is "1.235" then enter the answer as "1.24")

[Source ans: 0.40 to 0.42]

**12.** [Source Q17; SA]
If a randomly selected bulb from the production lot has a life of more than 100 hours, then what is the probability that it was of Type-B? (Note: Input your answer rounded to two decimal places. For example, if your answer is "1.235" then enter the answer as "1.24")

[Source ans: 0.28 to 0.30]

**13.** [Source Q18; SA]
DMS is giving a replacement warranty for bulbs that last for less than 100 hours of life. The replacement will cost Rs. 2 for every Type-A bulb, Rs. 1.75 for every Type-B bulb and Rs. 3 for every Type-C bulb. Then, if 10,000 bulbs are produced and sold in a year, what would be the total warranty related cost that DMS will expect to incur? (Note: Input your answer rounded to two decimal places. For example, if your answer is "1.235" then enter the answer as "1.24")

[Source ans: 14849 to 14851]

# 2024 Oct27 Quiz-1

**Comprehension for questions 14-18** (source Q3)

MSS is a firm that specializes in social media analytics. MSS has acquired data (in Table-1) on the past movie releases of two Heroes nick-named "Manda Kasayam (MK)" and "Youth Vodhavaka (YV)" in the Tamil Film industry. The data is on the Box-Office collections for the movies of MK and YV on different dates in January-1950. Then answer the given subquestions,with the presumption that the other necessary conditions required for the questions are met.

| Date | Day | Box-Office collection for MK (in INR Million) | Box-Office Collection for YV (in INR Million) |
|---|---|---|---|
| 01-Jan-1950 | Sunday | 11 | 10 |
| 02-Jan-1950 | Monday | 9 | 10 |
| 03-Jan-1950 | Tuesday | 15 | 9 |
| 04-Jan-1950 | Wednesday | 8 | 6 |
| 05-Jan-1950 | Thursday | 18 | 2 |
| 06-Jan-1950 | Friday | 16 | 10 |
| 07-Jan-1950 | Saturday | 19 | 7 |
| 08-Jan-1950 | Sunday | 8 | 6 |
| 09-Jan-1950 | Monday | 10 | 1 |
| 10-Jan-1950 | Tuesday | 19 | 3 |
| 11-Jan-1950 | Wednesday | 19 | 5 |
| 12-Jan-1950 | Thursday | 19 | 8 |
| 13-Jan-1950 | Friday | 15 | 8 |
| 14-Jan-1950 | Saturday | 17 | 5 |
| 15-Jan-1950 | Sunday | 15 | 2 |

Table-1

**14.** [Source Q4; SA]
The aim is to explore if the total box-office collection during the weekends (Friday, Saturday and Sunday) and the total box-office collection during the weekdays (Monday to Thursday) are independent across the two heroes. Then how many DOFs does the test have? (Note: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")

[Source ans: 1]

**15.** [Source Q5; SA]
The aim is to explore if the total box-office collection during the weekends (Friday, Saturday and Sunday) and the total box-office collection during the weekdays (Monday to Thursday) are independent across the two heroes. Then what is the expected box-office collection for MK ON WEEKENDS? (Note: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")

[Source ans: 103.00 to 105.00]

**16.** [Source Q6; SA]
The aim is to explore if the total box-office collection during the weekends (Friday, Saturday and Sunday) and the total box-office collection during the weekdays (Monday to Thursday) are independent across the two heroes. Then what is the expected box-office collection for YV ON WEEKDAYS? (Note: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")

[Source ans: 46.00 to 48.00]

**17.** [Source Q7; SA]
The aim is to explore if the total box-office collection during the weekends (Friday, Saturday and Sunday) and the total box-office collection during the weekdays (Monday to Thursday) are independent across the two heroes. Then what is value for the computed test statistic? (Note: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")

[Source ans: 0.85 to 0.92]

**18.** [Source Q8; MSQ]

The aim is to explore if the total box-office collection during the weekends (Friday, Saturday and Sunday) and the total box-office collection during the weekdays (Monday to Thursday) are independent across the two heroes. Then given the below Table-2, at a 95% confidence level, what is the conclusion?

| $\nu$ \ $\alpha$ | .995 | .990 | .975 | .950 | .900 | .500 | .100 | .050 | .025 | .010 | .005 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | .00+ | .00+ | .00+ | .00+ | .02 | .45 | 2.71 | 3.84 | 5.02 | 6.63 | 7.88 |
| 2 | .01 | .02 | .05 | .10 | .21 | 1.39 | 4.61 | 5.99 | 7.38 | 9.21 | 10.60 |
| 3 | .07 | .11 | .22 | .35 | .58 | 2.37 | 6.25 | 7.81 | 9.35 | 11.34 | 12.84 |
| 4 | .21 | .30 | .48 | .71 | 1.06 | 3.36 | 7.78 | 9.49 | 11.14 | 13.28 | 14.86 |
| 5 | .41 | .55 | .83 | 1.15 | 1.61 | 4.35 | 9.24 | 11.07 | 12.83 | 15.09 | 16.75 |

(a) Do Not Reject the Null

(b) Reject the Null

(c) Accept the Null

(d) Do Not Accept the Null

(e) None of these

[Source ans: a]

**Comprehension for questions 19-22** (source Q16)

You want to apply for a student visa to country "X". You can do this through any one of the five application centres, "A", "B", "C", "D" or "E". To determine which application centre to choose, you collect data. Currently, the embassy has decided to receive 15% of the applications from centre A, 10% from centre B, 25% each from centres C, D and E. Historically, 60% of the applications that come from centre A have been granted the visa, and 80% of the applications from centre B have been granted the visa. Centres C, D and E each had a 50-50 chance of granting a visa or not granting a visa respectively. Then answer the given subquestions.

**19.** [Source Q17; SA]
What is your probability of applying for a student visa through centre A and it is granted? (Note: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")

[Source ans: 0.09 to 0.10]

**20.** [Source Q18; SA]
What is your probability of applying for a student visa through centre B and it is not-granted? (Note: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")

[Source ans: 0.01 to 0.03]

**21.** [Source Q19; SA]
If a total of 500 applications have been granted a visa, then how many would you expect to have come from centres "C", "D" and "E" (totally from the three centres)? (Note: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")

[Source ans: 187 to 188]

**22.** [Source Q20; SA]
Say you find Dr.MS had applied for a student visa in the past, and the application was not granted. However, Dr.MS has not told you which application centre was chosen to submit the application. Then what is the probability that Dr. MS has applied though centre C? (Note: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")

[Source ans: 0.26 to 0.28]

## 2024 Dec22 ET

**23.** [Source Q2; SA]
Two factories "A" and "B" produce radios. Each radio produced at factory-A is defective with probability 0.05, whereas each radio produced at factory-B is defective with probability 0.1. You purchase two radios that were produced in the same factory (which is equally likely to have been either factory-A or factory-B). If the first radio that you check is defective, what is the conditional probability that the second radio is also defective? (Note: Enter your answer in decimals rounded to two decimal places. For example, if your answer is "0.456" then enter it as "0.46" )

[Source ans: 0.04 to 0.05]

**Comprehension for questions 24-26** (source Q8)

The BS program coordinators wanted to understand the difficulty level of the BA course. Hence, past learners of the BA course were asked to give a rating (on a scale of 1 to 5, 1 meaning "least difficult" and 5 implying "most difficult"). A total of 200 students (100 male and 100 female) were approached for the survey and the result of the same is provided in Table-2. Given this data, answer the sub-questions.

| Level of Difficulty for BA Course | Number of "Male Students" who opted for the level | Number of "Female Students" who opted for the level |
|---|---|---|
| 1 | 15 | 30 |
| 2 | 10 | 10 |
| 3 | 20 | 15 |
| 4 | 40 | 20 |
| 5 | 15 | 25 |

Table-2

**24.** [Source Q9; SA]
If the aim is to determine if the difficulty level is independent of gender, then how many male responses would you expect to choose the rating value of "3"? (Note: Enter your answer in decimal value rounded to two decimal places. For example, if your answer is "123.456" then enter it as "123.46" )

[Source ans: 17.00 to 18.00]

**25.** [Source Q10; SA]
If the aim is to determine if the difficulty level is independent of gender, then how many female responses would you expect to choose the rating value of "5"? (Note: Enter your answer in decimal value rounded to two decimal places. For example, if your answer is "123.456" then enter it as "123.46" )

[Source ans: 20.00]

**26.** [Source Q11; SA]
If the aim is to find if the difficulty level is independent of gender, then what is the value for the test statistic? (Enter your answer in decimals rounded to two decimal places. Eg: If your answer is 10.256, then enter it as 10.26 )

[Source ans: 14.50 to 15.00]

# 2025 Feb23 Quiz-1

**Comprehension for questions 27-31** (source Q4)

**(The following comprehension is purely imaginary)**

A restaurant ("Milo's Kitchen") has data pertaining to customers ordering three types of starters "Paneer Tikka (PT)", "Baby Corn Manchurian (BCM)" and "Masala Papad (MP)". For simplicity assume that the orders for starters is mutually exclusive and that a customer does not order more than one starter. Moreover, the customers who order these starters are predominantly one of the three types of eaters, "Roti Eaters", "Poori Eaters" or "Rice Eaters" ***(Note: customers can be "other eaters" as well. Hence, the list is not exhaustive)***. From past footfall data, it is seen that 25% of the customer's order "BCM" and another 40% order "PT". The remaining customers order "MP". It is observed that,

a. Of the people ordering "BCM", 30% are "Roti Eaters" and 60% are "Poori Eaters".
b. For "PT" orders, it is observed that 40% are "Rice Eaters", 20% are "Poori Eaters" and another 20% are "Roti Eaters".
c. 50% of the customers who order "MP" are "Rice Eaters" and 40% are "Poori Eaters"
d. "BCM" is never ordered by "Rice Eaters" and "MP" is never opted for by "Roti Eaters"

Assuming any other necessary mutual exclusiveness, and answer the given sub questions.

**27.** [Source Q5; SA]
If 1,000 customers come to the restaurant, how many will/may be "Rice Eaters"? (Note: Enter the answer rounded to two decimal places. For example, if the answer is "10.235", then enter it as "10.24". DO NOT ENTER PERCENTAGE VALUES)

[Source ans: 335]

**28.** [Source Q6; SA]
If 4,000 customers had visited the shop on 10th December 2024, then how many "BCMs" will you expect to serve? (Note: Enter the answer rounded to two decimal places. For example, if the answer is "10.235", then enter it as "10.24". DO NOT ENTER PERCENTAGE VALUES)

[Source ans: 1000]

**29.** [Source Q7; SA]
Milo's Kitchen is planning spend money on purchasing a refrigerator (for "Paneer" and "Baby Corn" storage). If the 600 "Poori Eaters" are expected to arrive, then what is the expected revenue generated from them if the restaurant charges Rs. 350 per PT order, Rs. 250 per BCM order and Rs. 80 per MP order? (Note: Enter the answer "in Rupees" rounded to two decimal places. For example, if the answer is "Rs. 10.235", then enter it as "10.24". DO NOT ENTER ANY SYMBOLS OR LETTERS)

[Source ans: 46020]

**30.** [Source Q8; SA]
If a total of 10,000 customers visit Milo's Kitchen in a month, how many (in Total) will you expect to be non-Rice Eater, Non-Poori Eater and Non-Roti Eater? (Note: Enter the answer rounded to two decimal places. For example, if the answer is "10.235", then enter it as "10.24". DO NOT ENTER PERCENTAGE VALUES)

[Source ans: 1400]

**31.** [Source Q9; MCQ]
Recently, the owner found a CCTV footage of a customer stealing a starter (but was not able to see which starter was stolen). The owner was sure that the food stealing customer was not a "Rice Eater", "Poori Eater" or "Roti Eater" . Then which starter would the owner suspect the stealing customer to have stolen?

(a) PT

(b) BCM

(c) MP

(d) None of these as Milo's Kitchen owner does not have a working CCTV (P.S: Thisis not an answer so don't choose this option. Just relax, think and answer the question)

[Source ans: a]

**Comprehension for questions 32-34** (source Q20)

**(The following comprehension is purely imaginary)**

You are given the following contingency table based on a sample data where different cities and their brand preferences are provided.

| | Brand A | Brand B | Total |
|---|---|---|---|
| Chennai | 200 | 145 | 345 |
| Mumbai | 600 | 255 | 855 |
| Total | 800 | 400 | 1200 |

Then answer the given subquestions.

**32.** [Source Q21; SA]
From the given contingency table, find the expected frequency of Chennai people preferring brand B? (Note: give only a numerical value rounded to 2 decimal places. For example, if your answer is 10.1253, then input the answer as "10.13")

[Source ans: 115]

**33.** [Source Q22; SA]
What is the calculated value of chi-squared? (Note: give only a numerical value rounded to 2 decimal places. For example, if your answer is 10.1253, then input the answer as "10.13")

[Source ans: 65.4 to 65.8]

**34.** [Source Q23; MSQ]
At the significance level 0.05, chi-squared tabular value is 3.84. What do you conclude? (choose all that is/are applicable)

(a) Reject the null hypothesis and conclude that the categorical variables are notindependent

(b) Fail to reject the null hypothesis and conclude that the categorical variablesare not independent

(c) Reject the null hypothesis and conclude that the categorical variables areindependent

(d) Fail to reject the null hypothesis and conclude that the categorical variablesare independent

(e) Accept the null hypothesis and conclude that the categorical variables are notindependent

(f) Do not accept the null hypothesis and conclude that the categorical variablesare not independent

(g) Accept the null hypothesis and conclude that the categorical variables areindependent

(h) Do not accept the null hypothesis and conclude that the categorical variablesare independent

(i) None of these

[Source ans: a]

## 2025 Mar16 Quiz-2

**35.** [Source Q2; SA]
The probability that a car battery functions for more than 10,000Kms is 0.8, the probability that it functions for more than 20,000kms is 0.4 and the probability that it functions for more than 30,000kms is 0.1. If a new car battery is still working after 10,000Kms, what is the probability that its total life will exceed 20,000Kms? (Note: Enter the answer rounded to two decimal places. For example, if the answer is "0.235", then enter it as "0.24")

[Source ans: 0.50]

# 2025 Apr13 ET

**Comprehension for questions 36-37** (source Q34)

(The following is a purely imaginary scenario) An insurance company believes that people can be divided into two classes: "Class-1: Those who are accident prone" and "Class-2: Those who are not accident prone". The company's statistics show that an accident-prone person will have an accident at sometime within a faxed 1-year period with probability 0.4, whereas this probability decreases to 0.2 for a person who is not accident prone. It is assumed that 30 percent of the human population is accident prone. Given this information, answer the sub-questions

**36.** [Source Q35; SA]
What is the probability that a new policyholder will have an accident within a year of purchasing a policy? (Note: Enter the answer rounded to two decimal places. For example, if the answer is "1.234", then enter it as "1.23" )

[Source ans: 0.25 to 0.27]

**37.** [Source Q36; SA]
Suppose a new policyholder has an accident within a year of purchasing a policy. Then, what is the probability that the person is accident prone? (Note: Enter the answer rounded to two decimal places. For example, if the answer is "1.234", then enter it as "1.23" )

[Source ans: 0.44 to 0.48]

## 2025 Jul13 Quiz-1

**38.** [Source Q5; SA]
Four teams (1, 2, 3 and 4) have qualified for the final of the cricket world cup. Team 1 plays Team 2. Team 3 plays Team 4. The winners then play in the final. In a game between Team "i" and Team "j", the probability of team "i" winning is given by {i /(i+j)}. Then what is the conditional probability that "Team 1" will win the World Cup? (Note: Enter your answer in decimals rounded to two decimal places. For example, if your answer is "0.245", enter it as "0.25" )

[Source ans: 0.06 to 0.09]

**Comprehension for questions 39-43** (source Q13)

A cosmetics brand "SwamiIsBeautiful" manufactures three products, "Face Creams (FC)", "Body Lotion (BL)" and "Lip Gloze (LG)". The company believes displaying all their products together in a single shelf at a retailer store will help boost their sales, as people tend to pick up the items together. Hence, they are currently spending a lot of money to make sure that the retailers display the products together. However, on hearing this, Dr.Milo feels sceptical as it is counter intuitive. He feels that the sales of the products are independent. To check his theory, data in Table-3 was collected. Table-3 contains the total sales (in quantity) of the three products across retail stores. Here, the retail stores were classified in two types: "Type-1: The three products are displayed together", "Type-2: The three products are not displayed together". Given this information answer the given subquestions. (Note: For all questions in this comprehension, if your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")

| Product | Retail Store Type | |
|---|---|---|
| | Type-1 | Type-2 |
| FC | 288 | 124 |
| BL | 622 | 204 |
| LG | 212 | 128 |

Table-3

**39.** [Source Q14; MCQ]
What type of representation is Table-3?

(a) Contingency Table

(b) Frequency Table

(c) None of these

(d) Both Contingency Table & Frequency Table

[Source ans: a]

**40.** [Source Q15; SA]
What is the expected frequency for "BL" in Type-2 stores?

[Source ans: 237 to 239]

**41.** [Source Q16; SA]
What is the value of the computed test statistic for the hypothesis test that DrMilo will perform to check if his intuition is correct?

[Source ans: 18 to 22]

**42.** [Source Q17; MSQ]
What is the Null Hypothesise for the hypothesis test that DrMilo will be performing? (choose all that is applicable)

(a) The total sales is independent among the different products and differentretail store types

(b) The total sales is not independent among the different products and differentretail store types

(c) The total sales is uniformly distributed among the different products anddifferent retail store types

(d) The total sales is uniformly distributed only among the different products

(e) The total sales is uniformly distributed only among the different retail storetypes

(f) None of these

[Source ans: a]

**43.** [Source Q18; MSQ]
Given Figure-2, at a 20% level of significance, what will be the conclusion of the hypothesis test? (choose all that is applicable)

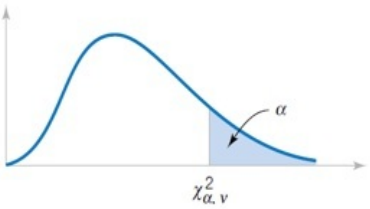



**Table III** Percentage Points $\chi^2_{\alpha,\nu}$ of the Chi-Squared Distribution

| $\nu$ \ $\alpha$ | .995 | .990 | .975 | .950 | .900 | .500 | .100 | .050 | .025 | .010 | .005 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | .00+ | .00+ | .00+ | .00+ | .02 | .45 | 2.71 | 3.84 | 5.02 | 6.63 | 7.88 |
| 2 | .01 | .02 | .05 | .10 | .21 | 1.39 | 4.61 | 5.99 | 7.38 | 9.21 | 10.60 |
| 3 | .07 | .11 | .22 | .35 | .58 | 2.37 | 6.25 | 7.81 | 9.35 | 11.34 | 12.84 |
| 4 | .21 | .30 | .48 | .71 | 1.06 | 3.36 | 7.78 | 9.49 | 11.14 | 13.28 | 14.86 |
| 5 | .41 | .55 | .83 | 1.15 | 1.61 | 4.35 | 9.24 | 11.07 | 12.83 | 15.09 | 16.75 |
| 6 | .68 | .87 | 1.24 | 1.64 | 2.20 | 5.35 | 10.65 | 12.59 | 14.45 | 16.81 | 18.55 |
| 7 | .99 | 1.24 | 1.69 | 2.17 | 2.83 | 6.35 | 12.02 | 14.07 | 16.01 | 18.48 | 20.28 |
| 8 | 1.34 | 1.65 | 2.18 | 2.73 | 3.49 | 7.34 | 13.36 | 15.51 | 17.53 | 20.09 | 21.96 |
| 9 | 1.73 | 2.09 | 2.70 | 3.33 | 4.17 | 8.34 | 14.68 | 16.92 | 19.02 | 21.67 | 23.59 |
| 10 | 2.16 | 2.56 | 3.25 | 3.94 | 4.87 | 9.34 | 15.99 | 18.31 | 20.48 | 23.21 | 25.19 |

*Figure-2*

(a) Reject the Null Hypothesis

(b) Do Not Reject the Null Hypothesis

(c) Reject the Alternative Hypothesis

(d) Do Not Reject the Alternative Hypothesis

(e) Accept the Null Hypothesis

(f) Do Not Accept the Null Hypothesis

(g) Accept the Alternative Hypothesis

(h) Do Not Accept the Alternative Hypothesis

[Source ans: a]

# 2025 Oct26 Quiz-1

**Comprehension for questions 44-46** (source Q4)

(The following is a purely imaginary scenario) An insurance company believes that people can be divided into two classes: "Class-1: Those who are accident prone" and "Class-2: Those who are not accident prone". The company's statistics show that an accident-prone person will have an accident at sometime within a fixed 1-year period with probability 0.4, whereas this probability decreases to 0.2 for a person who is not accident prone. It is assumed that 30 percent of the human population is accident prone. Given this information, answer the sub-questions.

**44.** [Source Q5; SA]
What is the probability that a new policyholder will have an accident within a year of purchasing a policy? (Note: Enter the answer rounded to two decimal places . For example, if the answer is "1.235", then enter it as "1.24" )

[Source ans: 0.25 to 0.27]

**45.** [Source Q6; SA]
Suppose you learn that a new policyholder has an accident within a year of purchasing a policy. Then, what is the probability that the person is accident prone? (Note: Enter the answer rounded to two decimal places . For example, if the answer is "1.235", then enter it as "1.24")

[Source ans: 0.44 to 0.48]

**46.** [Source Q7; SA]
The insurance company has to pay an insurance fee of Rs. 10 per accident if the accident happens in the first year. If the company has sold 1000 insurances in total, then how much money (in Rs.) will it expect to pay at the end of the first year for Class-2 people? (Note: Enter the answer rounded to two decimal places . For example, if the answer is "Rs. 1.235", then enter it as "1.24")

[Source ans: 1400]

**Comprehension for questions 47-49** (source Q17)

(The following is a purely imaginary scenario) A company sells two products (A and B) in four different marketplaces. In the first marketplace, the sales are made at the Forenoon (FN). In the second marketplace, sales are made during the Afternoon (AN). In the third marketplace, sales are made during the Evening (E). In the fourth marketplace, sales are made during the Night (N). Table-2 provides the data on the sales in the four marketplaces for the two products in the first 5 days on January 2023. Using this data, answer the given subquestions.

| Date | Marketplace | Number of units of Product-A Sold | Number of units of Product-B Sold |
|---|---|---|---|
| 01-01-2023 | FN | 31 | 30 |
| 02-01-2023 | FN | 21 | 41 |
| 03-01-2023 | FN | 16 | 45 |
| 04-01-2023 | FN | 7 | 56 |
| 05-01-2023 | FN | 13 | 49 |
| 01-01-2023 | AN | 8 | 40 |
| 02-01-2023 | AN | 8 | 42 |
| 03-01-2023 | AN | 21 | 44 |
| 04-01-2023 | AN | 13 | 55 |
| 05-01-2023 | AN | 7 | 42 |
| 01-01-2023 | E | 31 | 30 |
| 02-01-2023 | E | 21 | 41 |
| 03-01-2023 | E | 16 | 52 |
| 04-01-2023 | E | 7 | 57 |
| 05-01-2023 | E | 13 | 49 |
| 01-01-2023 | N | 8 | 55 |
| 02-01-2023 | N | 8 | 57 |
| 03-01-2023 | N | 21 | 44 |
| 04-01-2023 | N | 13 | 52 |
| 05-01-2023 | N | 7 | 62 |

Table-2

**47.** [Source Q18; SA]

The question of whether total sales in different marketplaces for product A and total sales in different marketplaces for product B are independent across the different days is being checked. Then what is the expected frequency for "Product-B" on "03-01-2025"? (Note: Enter the answer rounded to two decimal places . For example, if the answer is "1.235", then enter it as "1.24" )

[Source ans: 197.00 to 199.00]

**48.** [Source Q19; SA]
The question of whether total sales in different marketplaces for product A and total sales in different marketplaces for product B are independent across the different days is being checked. Then what is the value of the computed test statistic? (Note: Enter the answer rounded to two decimal places . For example, if the answer is "1.235", then enter it as "1.24")

[Source ans: 31.5 to 33.5]

**49.** [Source Q20; SA]
The question of whether total sales in different marketplaces for product A and total sales in different marketplaces for product B are independent across the different days is being checked. Then what is the number of degrees of freedom for the test? (Note: Enter the answer rounded to two decimal places . For example, if the answer is "1.235", then enter it as "1.24" )

[Source ans: 4]

# 2026 Mar15 Quiz-1

**Comprehension for questions 50-52** (source Q8)

Say Table 2, provides the data on the number of enrolments who complete the "Foundation", "Diploma", and "Degree" and the number of enrolments who "discontinue" the IITM BSc program for "Male" and "Female" gender students.

| | *Completed Foundation* | *Completed Diploma* | *Completed Degree* | *Discontinued* |
|---|---|---|---|---|
| ***Male*** | 55 | 45 | 20 | 53 |
| ***Female*** | 85 | 25 | 45 | 25 |

Table-2

Based on the above data, answer the given subquestions.

**50.** [Source Q9; SA]
If the aim is to understand if the "Gender" categories and "Completion" categories (including completion of any degree and discontinuation) are independent of each other, then what is the value for the computed test statistic? (Note: round your final answer to two decimal places. Example: If your answer is "1.2354" enter it as "1.24")

[Source ans: 31.00 to 32.00]

**51.** [Source Q10; SA]
What is the number of degrees of freedom for the statistical test to be performed? (Note: give only a numerical value rounded to 2 decimal places. For example, if your answer is 10.1253, then input the answer as "10.13")

[Source ans: 3.00]

**52.** [Source Q11; MSQ]
If a 20% significance level is considered, then which of the following statements is/are true for the performed statistical test (choose all that are applicable)

(a) If the p-value for the test is 0.25, conclude that the course completions are independent of gender

(b) If the p-value for the test is 0.15, conclude that the course completions are not independent of gender

(c) If the p-value for the test is 0.25, conclude that the course completions are not independent of gender

(d) If the p-value for the test is 0.15, conclude that the course completions are independent of gender

(e) If the computed test statistic is less than the chi-square tabulated, then conclude that the course enrolments are independent of gender

(f) If the computed test statistic is greater than the chi-square tabulated, then conclude that the course enrolments are not independent of gender

(g) If the computed test statistic is less than the chi-square tabulated, then conclude that the course enrolments are not independent of gender

(h) If the computed test statistic is greater than the chi-square tabulated, then conclude that the course enrolments are independent of gender

[Source ans: a, b, e, f]

**Comprehension for questions 53-55** (source Q12)

DMS lighting produces 3 types of bulbs (Type-A, Type-B and Type-C). The probability that a Type-A bulb will give more than 100 hours (>= 100 hours) of life is 0.8, with the corresponding probabilities of Type-B and Type-C bulbs being 0.6 and 0.7 respectively. In any production, 30% of the bulbs are of Type-A, 35% are of Type-B and 35% are of Type-C. Based on the above data, answer the given subquestions.

**53.** [Source Q13; SA]
What is the probability that a bulb from the production lot will give a life of more than 100 hours? (Note: Input your answer rounded to two decimal places. For example, if your answer is "1.235" then enter the answer as "1.24")

[Source ans: 0.65 to 0.71]

**54.** [Source Q14; SA]
If a randomly selected bulb from the production lot has a life of more than 100 hours, then what is the probability that it was of Type-C? (Note: Input your answer rounded to two decimal places. For example, if your answer is "1.235" then enter the answer as "1.24")

[Source ans: 0.33 to 0.37]

**55.** [Source Q15; SA]
DMS is giving a replacement warranty for bulbs that last for less than 100 hours of life. The replacement will cost Rs. 8 for every Type-A bulb, Rs. 5 for every Type-B bulb and Rs. 6 for every Type-C bulb. Then, if 2,00,000 bulbs are produced and sold in a year, what would be the total warranty related cost that DMS will expect to incur? (Note: Input your answer in Rupees rounded to two decimal places. For example, if your answer is "Rupees 100.235" then enter the answer as "100.24")

[Source ans: 361990 to 362010]

# 2026 May06 ET

**Comprehension for questions 56-57** (source Q49)

The following scenario is purely hypothetical Suppose a factory manufactures products on three machines A, B and C. Suppose 45% of total output comes from machine A, 35% of total output comes from machine B and 20% of total output comes from machine C. From the past data, it is known that 3% of products by machine A are defectives, 3% of products by machine B are defectives and 5% of products by machine C are defectives. Then answer the given subquestions.

**56.** [Source Q50; SA]
If Rs. 40 is incurred for the repair work of every defective product, then what would be the total expected expenditure towards repairs for a production lot of 10,000 products? (Note: Enter your answer "Rs" denomination rounded to two decimal places. For example, if your answer is "Rs. 1.235", then enter the answer as "1.24")

[Source ans: 136000]

**57.** [Source Q51; SA]
What is the probability that the product has come from machine B given that it is defective? (Note: Enter your answer in decimal values rounded to two decimal places. For example, if your answer is "1.235", then enter the answer as "1.24")

[Source ans: 0.28 to 0.32]

# 2026 Jul16 Quiz-1

**Comprehension for questions 58-60** (source Q6)

(The following comprehension is purely imaginary) The BA team is known for their clumsy QP setting. Hence, after every quiz the enrolled learners always raise concerns. These concerns can be grouped into five categories: "Category-1: Not a valid concern- No Correction", "Category-2: Valid Concern-Minor Correction", "Category-3: Valid Concern-Major Corrections", "Catgeory-4: Valid Concern-Open Marks" and "Catgeory-5: Those who claim marks for their misconception". From the past six terms (Jan-20 to Sep-21), data on the number of requests in each category is captured and presented in Table-2 below. Given this information, answer the given sub-questions

| | Jan-20 | May-20 | Sep-20 | Jan-21 | May-21 | Sep-21 | ***Row Total*** |
|---|---|---|---|---|---|---|---|
| Category-1 | 1 | 2 | 1 | 2 | 2 | 2 | *10* |
| Category-2 | 5 | 2 | 2 | 0 | 2 | 4 | *15* |
| Category-3 | 2 | 6 | 6 | 5 | 5 | 1 | *25* |
| Category-4 | 1 | 2 | 1 | 2 | 2 | 2 | *10* |
| Category-5 | 10 | 12 | 4 | 3 | 3 | 8 | *40* |
| ***Column Total*** | *19* | *24* | *14* | *12* | *14* | *17* | |

Table-2: Category-Term Count Data

**58.** [Source Q7; SA]
If the aim is to check if the counts of observations in each category is independent across terms, then what is the value for the expected frequency for the combination "Category-3 and Sep-20"? (Note: Enter your answer in decimals rounded to two decimal places. For example, if your answer is "0.245", enter it as "0.25")

[Source ans: 3.30 to 3.70]

**59.** [Source Q8; SA]
If the aim is to check if the counts of observations in each category is independent across terms, then what is the degrees of freedom for the test-statistic? (Note: Enter your answer in decimals rounded to two decimal places. For example, if your answer is "0.245", enter it as "0.25")

[Source ans: 19.90 to 20.10]

**60.** [Source Q9; MSQ]

If the aim is to check if the counts of observations in each category is independent across terms and say the value for the computed test statistic is 19.61, then based on Figure-1 what can be concluded? (Choose all that is/are applicable)

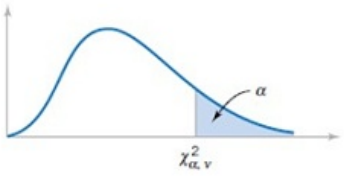



**Table III** Percentage Points $\chi^2_{\alpha,\nu}$ of the Chi-Squared Distribution

| ν \ α | .995 | .990 | .975 | .950 | .900 | .500 | .100 | .050 | .025 | .010 | .005 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | .00+ | .00+ | .00+ | .00+ | .02 | .45 | 2.71 | 3.84 | 5.02 | 6.63 | 7.88 |
| 2 | .01 | .02 | .05 | .10 | .21 | 1.39 | 4.61 | 5.99 | 7.38 | 9.21 | 10.60 |
| 3 | .07 | .11 | .22 | .35 | .58 | 2.37 | 6.25 | 7.81 | 9.35 | 11.34 | 12.84 |
| 4 | .21 | .30 | .48 | .71 | 1.06 | 3.36 | 7.78 | 9.49 | 11.14 | 13.28 | 14.86 |
| 5 | .41 | .55 | .83 | 1.15 | 1.61 | 4.35 | 9.24 | 11.07 | 12.83 | 15.09 | 16.75 |
| 6 | .68 | .87 | 1.24 | 1.64 | 2.20 | 5.35 | 10.65 | 12.59 | 14.45 | 16.81 | 18.55 |
| 7 | .99 | 1.24 | 1.69 | 2.17 | 2.83 | 6.35 | 12.02 | 14.07 | 16.01 | 18.48 | 20.28 |
| 8 | 1.34 | 1.65 | 2.18 | 2.73 | 3.49 | 7.34 | 13.36 | 15.51 | 17.53 | 20.09 | 21.96 |
| 9 | 1.73 | 2.09 | 2.70 | 3.33 | 4.17 | 8.34 | 14.68 | 16.92 | 19.02 | 21.67 | 23.59 |
| 10 | 2.16 | 2.56 | 3.25 | 3.94 | 4.87 | 9.34 | 15.99 | 18.31 | 20.48 | 23.21 | 25.19 |
| 11 | 2.60 | 3.05 | 3.82 | 4.57 | 5.58 | 10.34 | 17.28 | 19.68 | 21.92 | 24.72 | 26.76 |
| 12 | 3.07 | 3.57 | 4.40 | 5.23 | 6.30 | 11.34 | 18.55 | 21.03 | 23.34 | 26.22 | 28.30 |
| 13 | 3.57 | 4.11 | 5.01 | 5.89 | 7.04 | 12.34 | 19.81 | 22.36 | 24.74 | 27.69 | 29.82 |
| 14 | 4.07 | 4.66 | 5.63 | 6.57 | 7.79 | 13.34 | 21.06 | 23.68 | 26.12 | 29.14 | 31.32 |
| 15 | 4.60 | 5.23 | 6.27 | 7.26 | 8.55 | 14.34 | 22.31 | 25.00 | 27.49 | 30.58 | 32.80 |
| 16 | 5.14 | 5.81 | 6.91 | 7.96 | 9.31 | 15.34 | 23.54 | 26.30 | 28.85 | 32.00 | 34.27 |
| 17 | 5.70 | 6.41 | 7.56 | 8.67 | 10.09 | 16.34 | 24.77 | 27.59 | 30.19 | 33.41 | 35.72 |
| 18 | 6.26 | 7.01 | 8.23 | 9.39 | 10.87 | 17.34 | 25.99 | 28.87 | 31.53 | 34.81 | 37.16 |
| 19 | 6.84 | 7.63 | 8.91 | 10.12 | 11.65 | 18.34 | 27.20 | 30.14 | 32.85 | 36.19 | 38.58 |
| 20 | 7.43 | 8.26 | 9.59 | 10.85 | 12.44 | 19.34 | 28.41 | 31.41 | 34.17 | 37.57 | 40.00 |
| 21 | 8.03 | 8.90 | 10.28 | 11.59 | 13.24 | 20.34 | 29.62 | 32.67 | 35.48 | 38.93 | 41.40 |
| 22 | 8.64 | 9.54 | 10.98 | 12.34 | 14.04 | 21.34 | 30.81 | 33.92 | 36.78 | 40.29 | 42.80 |
| 23 | 9.26 | 10.20 | 11.69 | 13.09 | 14.85 | 22.34 | 32.01 | 35.17 | 38.08 | 41.64 | 44.18 |
| 24 | 9.89 | 10.86 | 12.40 | 13.85 | 15.66 | 23.34 | 33.20 | 36.42 | 39.36 | 42.98 | 45.56 |
| 25 | 10.52 | 11.52 | 13.12 | 14.61 | 16.47 | 24.34 | 34.28 | 37.65 | 40.65 | 44.31 | 46.93 |
| 26 | 11.16 | 12.20 | 13.84 | 15.38 | 17.29 | 25.34 | 35.56 | 38.89 | 41.92 | 45.64 | 48.29 |
| 27 | 11.81 | 12.88 | 14.57 | 16.15 | 18.11 | 26.34 | 36.74 | 40.11 | 43.19 | 46.96 | 49.65 |
| 28 | 12.46 | 13.57 | 15.31 | 16.93 | 18.94 | 27.34 | 37.92 | 41.34 | 44.46 | 48.28 | 50.99 |
| 29 | 13.12 | 14.26 | 16.05 | 17.71 | 19.77 | 28.34 | 39.09 | 42.56 | 45.72 | 49.59 | 52.34 |
| 30 | 13.79 | 14.95 | 16.79 | 18.49 | 20.60 | 29.34 | 40.26 | 43.77 | 46.98 | 50.89 | 53.67 |

Figure-1: Chi-Square Tabulated Values

(a) At a 95% confidence level, REJECT THE NULL and conclude that the categorical variables ARE INDEPENDENT

(b) At a 95% confidence level, REJECT THE NULL and conclude that the categorical variables ARE NOT INDEPENDENT

(c) At a 95% confidence level, DO NOT REJECT THE NULL and conclude that the categorical variables ARE INDEPENDENT

(d) At a 95% confidence level, DO NOT REJECT THE NULL and conclude that the categorical variables ARE NOT INDEPENDENT

(e) At a 50% confidence level, DO NOT REJECT THE NULL and conclude that the categorical variables ARE INDEPENDENT

(f) At a 50% confidence level, DO NOT REJECT THE NULL and conclude that the categorical variables ARE NOT INDEPENDENT

(g) At a 50% confidence level, REJECT THE NULL and conclude that the categorical variables ARE INDEPENDENT

(h) At a 50% confidence level, REJECT THE NULL and conclude that the categorical variables ARE NOT INDEPENDENT

(i) The p-value for the test is between 0.5 to 0.1 and more towards 0.1

(j) The p-value for the test is between 0.5 to 0.1 and more towards 0.5

(k) The p-value for the test is between 0.95 to 0.9 and more towards 0.9

(l) The p-value for the test is between 0.95 to 0.9 and more towards 0.95

(m) More information is required to compute the p-value

(n) A Typical Category-5 answer: As this was not covered in the course, it should be given open marks. (Note: This is not a correct answer so don't choose it. You will not be awarded marks if you choose and then try to claim marks through discourse)

[Source ans: c, h, i]

**Comprehension for questions 61-62** (source Q17)

(The following comprehension is purely imaginary) Suppose a factory manufactures products on three machines A, B and C. Suppose 65% of total output comes from machine A, 30% of total output comes from machine B and 5% of total output comes from machine C. From the past data, it is known that 1% of products by machine A are defectives, 2% of products by machine B are defectives and 10% of products by machine C are defectives. Then answer the given sub-questions

**61.** [Source Q18; SA]
What is the probability that the product has come from machine B, given that it is defective? (Note: Enter your answer in decimals rounded to two decimal places. For example, if your answer is "0.245", enter it as "0.25")

[Source ans: 0.32 to 0.36]

**62.** [Source Q19; SA]
If the company decides to provide a refund of Rs. 20 for a defective product, then what would be the expected total cost for refund if a total of 1000 products where made? (Note: Enter your answer in decimals rounded to two decimal places. For example, if your answer is "0.245", enter it as "0.25")

[Source ans: 349.00 to 351.00]